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Research Article

SMART Vet: A Knowledge Management Platform Application for Enhancing Local Government Unit Veterinary Services

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ABSTRACT

The SMART Vet knowledge management platform application represents a comprehensive digital solution designed to enhance veterinary service delivery within local government units. The research addresses critical inefficiencies in traditional veterinary management systems by developing an integrated platform featuring multi-platform accessibility, administrator dashboards, client registration systems, real-time messaging capabilities, appointment scheduling modules, and community animal reporting functions. The platform was systematically developed using the Waterfall Model methodology. The research employed a mixed-methods evaluation framework combining quantitative usability assessment through ISO/IEC 25010 software quality standards and qualitative stakeholder analysis via structured focus group discussions with veterinary professionals in San Fernando City, La Union. Results demonstrated exceptional system performance with an overall usability rating of 4.49 out of 5.0, officially classified as "Excellent and Very Highly Usable" according to ISO/IEC 25010 benchmarks. The platform exhibited particularly strong performance in Portability (4.67) and Performance Efficiency (4.59) metrics. Qualitative thematic analysis revealed unanimous professional endorsement across eight critical acceptability criteria, with participants identifying significant potential for improved service delivery and modernized operational workflows.
Despite positive reception, implementation challenges included inadequate technological infrastructure within government facilities, limited internet connectivity,

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and substantial ongoing IT support requirements. SWOT analysis confirmed strong market potential as the region's first veterinary management system while highlighting critical dependencies on infrastructure development and institutional partnerships. This research contributes to veterinary technology and e-governance knowledge by demonstrating the feasibility of digital veterinary solutions in resource-constrained environments, recommending phased implementation strategies for sustainable deployment.

Introduction

Government veterinarians play a vital role in safeguarding public health, promoting animal welfare, and ensuring food safety through implementing animal health regulations, managing zoonotic diseases, and supporting agriculture. However, the Philippines faces a critical challenge in veterinary service delivery that undermines these essential functions: fragmented data systems, inadequate animal health tracking, and reliance on manual processes that create significant gaps in disease surveillance and livestock management. This systemic problem extends beyond operational inefficiencies to impact public health outcomes and economic productivity, particularly as the country experiences steady livestock sector growth with the Cordillera Administrative Region producing 7,038 metric tons in 2024 and Central Visayas showing a 5.24% increase in livestock population. The absence of integrated digital veterinary platforms prevents effective coordination between local government units, hampers rapid response to zoonotic disease outbreaks, and limits the ability to implement evidence-based animal welfare policies.

This technological gap becomes particularly problematic within the context of the Philippines' broader digital governance transformation agenda. While Republic Act 7160 mandates that provinces and cities maintain veterinary officers responsible for developing and implementing veterinary service strategies, and Republic Act No. 11032 (Ease of Doing Business Act) requires government agencies to implement digital solutions to enhance service accessibility, the reality of veterinary service delivery remains largely analog. The disconnect between national digital transformation policies and local implementation capacity creates a critical void where the promise of e-governance meets the harsh reality of resource constraints and technological limitations. Poor data sharing between veterinary offices prevents coordinated responses to disease outbreaks, while weak animal health tracking systems compromise both livestock productivity and food safety monitoring, ultimately affecting economic growth and public health protection.

International organizations have demonstrated the transformative potential of technology-based veterinary systems, yet significant methodological and coverage gaps exist in current global platforms. The World Organization for Animal Health (WOAH) operates the World Animal Health Information System (WAHIS) for global disease surveillance, focusing primarily on epidemiologically important diseases in domestic and wild animals for international trade and policy decisions. Similarly, the Food and Agriculture Organization (FAO) utilizes EMPRES-I for real-time disease tracking and early warning alerts, emphasizing transboundary animal disease prevention and control. While these systems excel in large-scale surveillance and international coordination, they exhibit critical limitations in addressing local service delivery needs, community-level animal welfare management, and the specific regulatory requirements of individual countries.

Research demonstrates that countries using these international systems have achieved more effective outbreak management and improved response times, yet these platforms remain inadequate for local government operations requiring comprehensive service integration. WAHIS and EMPRES-i lack functionality for appointment scheduling, client communication, individual animal profile management, and local regulatory compliance—essential components for effective municipal veterinary services. Furthermore, these systems are designed for national and international coordination rather than direct citizen engagement, creating a significant gap in service accessibility for local communities. The technological architecture of existing platforms also presents challenges for resource-limited settings, often requiring extensive technical infrastructure and training that exceed the capacity of many local government units in developing countries.

Current literature reveals extremely limited technological advancement in Philippine LGU veterinary services, with the Las Piñas City Veterinary Office IT Management System, introduced in February 2023, representing the country's first and only LGU veterinary IT system. However, this system exemplifies the limitations of current approaches, remaining restricted to local area network access, requiring in-person visits for service completion, and focusing solely on pet registration through unique IDs and QR codes. This singular implementation highlights the substantial gap between international best practices and local implementation capacity, demonstrating how existing solutions fail to address the comprehensive needs of Philippine LGUs operating within specific legal frameworks and resource constraints.

The City of San Fernando, La Union, exemplifies this technological gap despite its commitment to becoming "A People's City in Region 1" through the San Fernando Tayo strategy focusing on Health, Economic Recovery, and Education. The city's veterinary office lacks any existing digital platform, continuing to rely entirely on traditional manual processes that limit efficient data collection, storage, and analysis capabilities essential for informed decision-making and improved service delivery. This situation reflects a broader pattern across Philippine LGUs where the absence of locally tailored digital tools prevents the realization of digital governance benefits, creating inefficiencies that compound over time and ultimately compromise service quality and public health outcomes.

SMART Vet emerges as a direct response to this multifaceted challenge, bridging the gap between the promise of digital transformation and the practical reality of local veterinary service delivery. Unlike existing international platforms that focus primarily on surveillance and monitoring, SMART Vet represents a flexible and locally-tailored solution that combines mobile data collection capabilities, integration with local veterinary practices, and alignment with Philippine regulatory frameworks. The platform addresses the unique challenges of resource-limited settings by providing comprehensive functionality through an integrated web-based and mobile platform that includes client registration with PRC ID verification for veterinarians, appointment scheduling with automated reminders, animal profile management with health monitoring, impounded animal tracking with adoption features, real-time messaging between clients and veterinary staff, and centralized administrative control—all designed to operate effectively within existing infrastructure constraints.

The platform's adaptive design philosophy shifts the academic discussion toward the unique value of locally responsive digital tools that can function effectively in resource-constrained environments while maintaining comprehensive functionality. SMART Vet incorporates

specific features addressing Philippine legal requirements, including compliance with Republic Act No. 8485 (Animal Welfare Act) through animal reporting and welfare monitoring capabilities, and local ordinances such as San Fernando's Anti-Stray Animal Ordinance through targeted impounded animal management features. This localization approach contrasts sharply with international platforms that require significant adaptation for local use, demonstrating how purpose-built solutions can more effectively serve community needs while supporting broader digital governance objectives.

The study's theoretical foundation integrates Public Choice Theory emphasizing efficiency and competition in government services, New Public Management principles focusing on customer-oriented governance and performance measurement, Technology Acceptance Model addressing perceived usefulness and ease of use, E-Government Theory promoting digital transformation and transparency, and Wiig's Knowledge Management Model organizing public, shared, and personal knowledge for optimal service delivery. This multi-theoretical approach ensures the platform addresses efficiency, user acceptance, knowledge management, and governance transformation simultaneously while remaining practically implementable within existing organizational and technological constraints.

SMART Vet contributes significantly to Sustainable Development Goals, particularly SDG 3 (Good Health and Well-Being) through comprehensive animal bite tracking and zoonotic disease prevention, SDG 8 (Decent Work and Economic Growth) by streamlining operations and supporting various stakeholders including farmers and livestock raisers, SDG 9 (Industry, Innovation, and Infrastructure) through technological advancement in veterinary service infrastructure, and SDG 17 (Partnerships for the Goals) by facilitating multi-stakeholder collaboration through integrated communication features. The research methodology employs the Waterfall development model with systematic usability evaluation across eight key areas: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability, complemented by stakeholder feedback collection through focus groups and pilot testing to ensure comprehensive assessment and continuous improvement. This research addresses the critical need for technological advancement in Philippine LGU veterinary services by developing a comprehensive, locally tailored solution that transcends the limitations of existing international platforms and current local implementations. Through systematic evaluation and stakeholder engagement, the study aims to provide a scalable model for nationwide adoption that demonstrates how adaptive digital tools can effectively bridge the gap between digital governance aspirations and practical implementation realities, ultimately improving animal welfare, public health protection, and government service efficiency across Philippine communities while contributing to global sustainable development objectives.

Materials and Methods

This study employed a mixed-methods approach combining developmental and descriptive research designs to create and evaluate the SMART Vet knowledge management platform application. The developmental research design facilitated the systematic creation and refinement of the platform through iterative testing, stakeholder feedback integration, and continuous improvement processes, following established developmental research methodologies that emphasize user-centered design and evidence-based enhancement (Wang & Zhang, 2020). The descriptive research design, incorporating both quantitative and qualitative methods, was utilized to comprehensively evaluate the application's usability across multiple software quality characteristics and assess its acceptability among diverse veterinary stakeholders through structured assessment protocols.

The research was conducted at the City Veterinarian Office of San Fernando, La Union, Philippines, following formal ethical approval and administrative clearance from the City Mayor's office. This location was strategically selected as it represents a typical Philippine LGU veterinary office that lacks existing digital platforms, making it an ideal setting for evaluating the practical implementation and acceptance of the SMART Vet system. The office provides comprehensive veterinary services including vaccination programs, clinical consultations, health certifications, artificial insemination services, meat inspection protocols, and livestock management programs, representing the full spectrum of services that the platform aims to digitize and enhance.

The SMART Vet platform was developed using the Waterfall Model, selected for its systematic approach to software development that ensures thorough documentation and quality control at each phase. The model was implemented through five sequential phases, each with specific deliverables and stakeholder involvement protocols. The Requirements Gathering phase involved extensive stakeholder consultations with LGU veterinarians, private veterinary practitioners, and service users to identify functional and non-functional requirements. Through structured interviews and observational studies of existing manual processes, the research team documented user needs, system constraints, and integration requirements with existing LGU workflows. This phase resulted in comprehensive requirement specifications that informed subsequent development decisions and ensured alignment with actual user needs and operational contexts.

The Design phase translated requirements into technical specifications and user interface mockups, employing user-centered design principles to ensure accessibility across diverse user groups with varying technological expertise. System architecture was designed to accommodate future scalability while maintaining performance efficiency within resource-limited environments typical of Philippine LGUs. Database schemas were developed to ensure data integrity and security while facilitating efficient information retrieval and reporting capabilities. The Implementation phase utilized a comprehensive technology stack optimized for cross-platform compatibility and performance. Flutter served as the primary development framework for mobile application development, selected for its ability to create high-performance, visually consistent applications across Android platforms using a single codebase. The Dart programming language was employed for application logic implementation, enabling efficient development with clean, maintainable code architecture. Visual Studio Code functioned as the primary integrated development environment (IDE), providing comprehensive debugging capabilities and code management features that enhanced development efficiency. Android Studio was utilized for Android-specific optimization and testing, ensuring compatibility across diverse Android device configurations commonly used in Philippine LGU settings. Data management utilized JavaScript Object Notation (JSON) for efficient data structuring and exchange between front-end and back-end components, facilitating seamless communication and maintaining data consistency across platform components. Web deployment infrastructure was established through Z.com domain registration services and dedicated web hosting subscriptions, ensuring reliable 24/7 accessibility with adequate bandwidth and server maintenance protocols to support concurrent user access typical of municipal government operations.

The Verification and Testing phase involved comprehensive quality assurance protocols including unit testing, integration testing, and user acceptance testing with representative stakeholder groups. Beta testing was conducted with selected LGU staff and community members to identify usability issues and gather preliminary feedback for system refinement.

The Deployment phase included system installation, user training protocols, and technical support framework establishment to ensure successful platform adoption and sustained usage. Purposive sampling was strategically employed to select participants based on Nielsen's (2000) established usability testing principles, which demonstrate that 5 users can identify approximately 80% of usability issues, while larger sample sizes provide diminishing returns in issue identification. The sampling strategy was designed to ensure comprehensive representation of the veterinary service ecosystem while maintaining methodological rigor and practical feasibility.

For usability assessment, 30 respondents were carefully selected representing six distinct stakeholder categories, with five participants from each group: veterinarians (both LGU and private sector), pet owners, farmers, livestock raisers, meat vendors, and animal transporters. This stratified sampling approach ensured representation of all primary user groups who would interact with the platform in real-world implementation scenarios. Participants were selected based on specific criteria including active engagement with veterinary services, technological literacy levels representative of the broader community, and willingness to provide comprehensive feedback throughout the testing process.

For acceptability assessment, 9 veterinarians were purposively selected comprising 2 from the LGU veterinary office and 7 from private practice, following optimal focus group discussion size recommendations established by Morgan (1998) and Krueger & Casey (2014). This composition ensured representation of both public and private sector perspectives while maintaining group dynamics conducive to meaningful discussion and comprehensive feedback generation. LGU veterinarians were selected based on their administrative roles, system implementation responsibilities, and extensive knowledge of current service delivery challenges. Private sector veterinarians were chosen to represent diverse practice types, technological adoption experiences, and client service approaches that could inform platform enhancement strategies.

Data collection employed multiple validated instruments and protocols designed to capture comprehensive usability and acceptability assessments. The ISO/IEC 25010 Product Quality Model served as the theoretical framework for usability evaluation, assessing eight critical software quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. This internationally recognized standard provided a systematic approach to software quality assessment that ensures comprehensive evaluation across all critical performance dimensions. A structured Google Forms questionnaire was developed and validated through pilot testing with a representative subset of participants before full deployment. The questionnaire employed a 5-point Likert scale with clearly defined descriptive anchors (1.00-1.79 = Not Usable; 1.80-2.59 = Fairly Usable; 2.60-3.39 = Usable; 3.40-4.19 = Highly Usable; 4.20-5.00 = Very Highly Usable) to enable precise measurement of participant perceptions across all quality dimensions. Each quality characteristic was assessed through multiple items to ensure measurement reliability and validity, with questions designed to capture both technical performance and user experience perspectives.

Focus Group Discussion protocols were developed following established qualitative research methodologies, incorporating both structured discussion topics and open-ended exploration opportunities. A hybrid approach was implemented with private sector veterinarians participating via Google Meet to accommodate scheduling constraints and geographic distribution, while LGU participants engaged in face-to-face discussions to enable direct

platform demonstration and hands-on interaction assessment. A trained technical assistant documented all interactions, navigated the application during demonstrations, and ensured consistent data capture across all sessions.

Thematic analysis was conducted following Braun & Clarke's (2006) six-phase framework: familiarization with data, initial code generation, theme identification, theme review, theme definition and naming, and report production. Multiple researchers independently coded transcripts to enhance reliability, with inter-coder agreement assessment and discussion of discrepancies to ensure analytical rigor. Coding strategies included both inductive and deductive approaches, allowing for identification of unexpected themes while ensuring systematic coverage of predetermined research questions.

SWOT analysis was employed as a complementary analytical framework to systematically categorize feedback into Strengths, Weaknesses, Opportunities, and Threats, providing strategic insights for platform enhancement and implementation planning. This structured approach facilitated the translation of qualitative feedback into actionable recommendations for system improvement and deployment strategy development.

Comprehensive ethical protocols were implemented throughout the research process to ensure participant protection and data security. Formal ethical approval was obtained from relevant institutional review boards, with additional administrative clearance secured from the City Government of San Fernando, La Union, ensuring compliance with both academic research standards and local government protocols for conducting research within municipal facilities. Informed consent procedures were rigorously implemented with all participants receiving detailed information about research objectives, data collection methods, potential risks and benefits, and their rights as research participants. Written consent was obtained from all participants, with clear explanation of voluntary participation principles and the right to withdraw from the study at any time without penalty. Special attention was given to ensuring informed consent comprehension across diverse educational backgrounds and technological literacy levels represented in the participant sample.

Data confidentiality and anonymity protocols were strictly maintained throughout all phases of data collection and analysis. No personal identifying information was collected through survey instruments, with all responses linked only to demographic categories necessary for analytical purposes. Focus group participants were assigned coded identifiers to protect individual identity while enabling analytical tracking of responses and themes. All audio recordings and transcripts were stored in encrypted formats with access limited to authorized research team members. Data usage was restricted exclusively to research purposes as explicitly stated in consent documents, with clear communication to participants about how their contributions would be used in research reporting and publication. All data presentation follows anonymization protocols that prevent identification of individual participants or organizations, ensuring that research findings contribute to scientific knowledge while protecting participant privacy and organizational confidentiality. Long-term data retention and destruction protocols comply with institutional guidelines and ethical standards for responsible research data management.

Results and Discussion

The SMART Vet knowledge management platform application represents a comprehensive digital transformation initiative for veterinary service delivery in the Philippines, developed using the Waterfall Model methodology through a systematic five-phase approach

encompassing requirements gathering, design, implementation, verification, and deployment. The platform successfully integrates modern technology with public veterinary services while maintaining strict compliance with Philippine legislation including the Ease of Doing Business Act, Data Privacy Act, and DICT Act, demonstrating alignment with the Department of Information and Communications Technology's vision of 'One Digitized Government' for the Philippines. The development process exemplified participatory digital governance by incorporating extensive stakeholder feedback through pilot testing and focus group discussions with both LGU and private veterinarians, establishing a foundation for operationalizing legislative mandates while enhancing transparency, accountability, and efficient service delivery in local government operations.

The platform features modern minimalist design principles with material design standards, ensuring clean interfaces and intuitive navigation across both mobile and web platforms through purple-gradient sections, card-based layouts, and responsive design that maintains brand consistency while optimizing user experience across devices. Multi-platform accessibility through both web and mobile interfaces eliminates traditional bureaucratic barriers, with sophisticated user management systems incorporating role-based access for administrators, veterinarians, and non-veterinarian clients, PRC license verification for veterinarian registration, and secure authentication processes that maintain professional standards while ensuring data privacy compliance. Core functionalities encompass integrated messaging systems facilitating direct communication between government service providers and citizens, appointment scheduling requiring LGU veterinarian approval, document management capabilities for ordinances and policies, interactive map-based veterinary station location services, comprehensive animal reporting systems, and sophisticated data management tools for animal profiles and impound records.

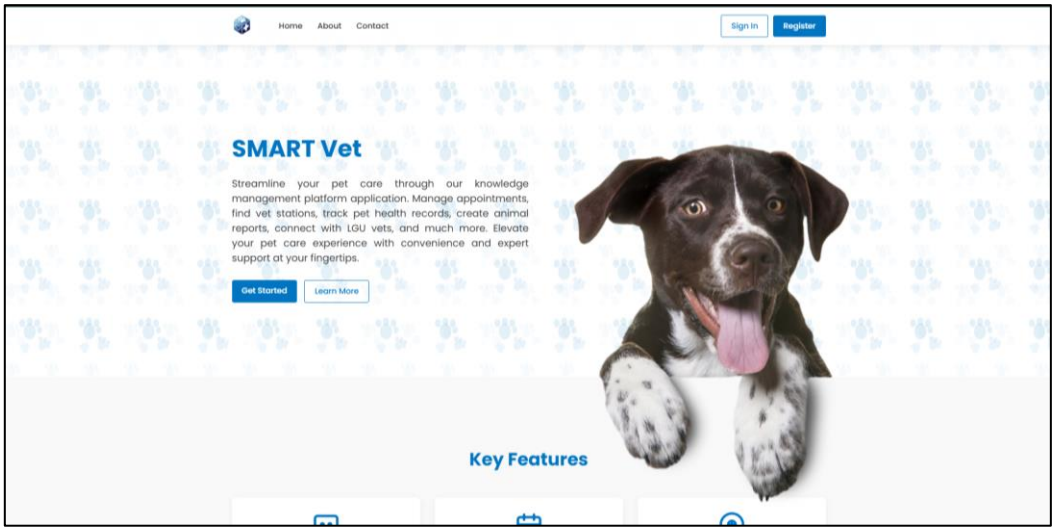

Plate 1. SMART Vet Web Version

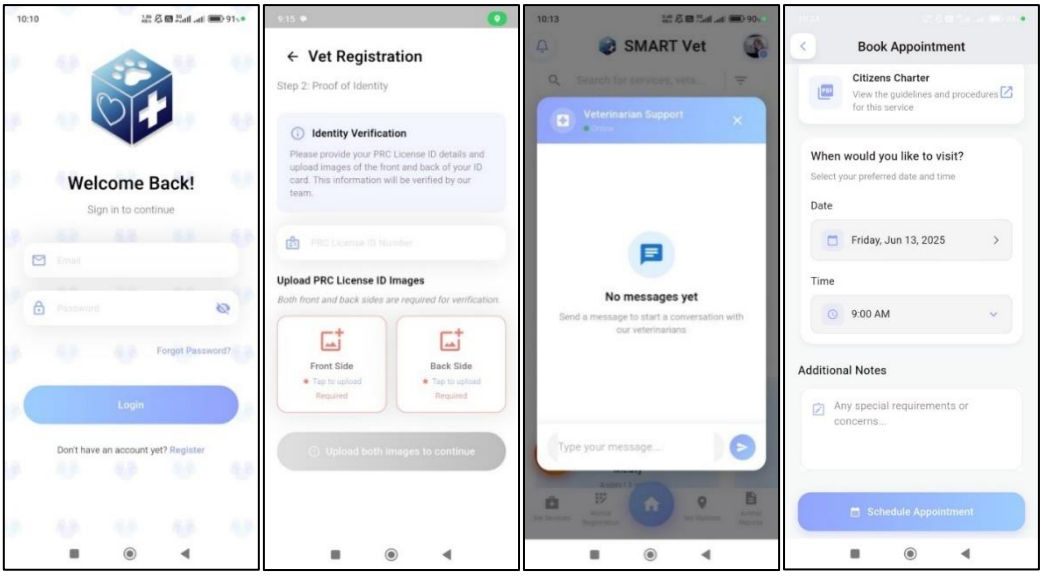

Plate 2. SMART Vet Mobile Version

The comprehensive usability evaluation demonstrated exceptional performance across all assessed quality characteristics, achieving an overall mean rating of 4.49 corresponding to an "Excellent" rating and "Very Highly Usable" interpretation according to established evaluation standards. Portability achieved the highest score of 4.67, indicating excellent adaptability across various platforms and environments, while Performance Efficiency scored 4.59, demonstrating superior operational effectiveness and responsiveness under different computational loads. Maintainability scored 4.51, reflecting excellent capabilities for system upkeep and modifications, while Functional Suitability achieved 4.50, confirming effective user requirement fulfillment and accurate performance of intended functions. Usability and Security both scored 4.43, with Compatibility achieving 4.40 and Reliability scoring 4.38, providing compelling evidence that SMART Vet consistently meets and exceeds established international software quality standards.

Table 1. Summary of Level of Usability of SMART Vet

Indicators	**Mean Rating**	**Descriptive Equivalent Rating**	**Descriptive Interpretation**
Functional Suitability	4.50	Excellent	Very Highly Usable
Performance Efficiency	4.59	Excellent	Very Highly Usable
Compatibility	4.40	Excellent	Very Highly Usable
Usability	4.43	Excellent	Very Highly Usable
Reliability	4.38	Excellent	Very Highly Usable
Security	4.43	Excellent	Very Highly Usable
Maintainability	4.51	Excellent	Very Highly Usable
Portability	4.67	Excellent	Very Highly Usable
Mean	4.49	Excellent	Very Highly Usable

The SMART Vet platform demonstrated strong user-centered design through systematic feedback integration during pilot testing, where user suggestions based on ISO/IEC 25010:2011 standards were analyzed, resulting in seven key recommendations including pet birthday notifications, "For Adoption" options, "Deceased" marking, appointment reminders, and app store uploads, with positive feedback on design usefulness and recommendations for city government adoption. Following evaluation, the researcher implemented four feasible enhancements: pet birthday notification features to enhance user engagement and support responsible pet ownership through proactive care reminders; "For Adoption" options within the Impound Animal feature to address animal welfare management gaps by facilitating rehoming processes; "Deceased" marking functionality for essential lifecycle management and accurate record-keeping; and appointment reminder sections to improve service accessibility and reduce missed appointments while enhancing LGU veterinary operation efficiency. These enhancements demonstrate how user feedback can be systematically incorporated through iterative development principles to create a more comprehensive and effective knowledge management platform that better serves both veterinary professionals and communities, validating the platform's responsiveness to user needs and commitment to continuous improvement based on evidence-based feedback integration.

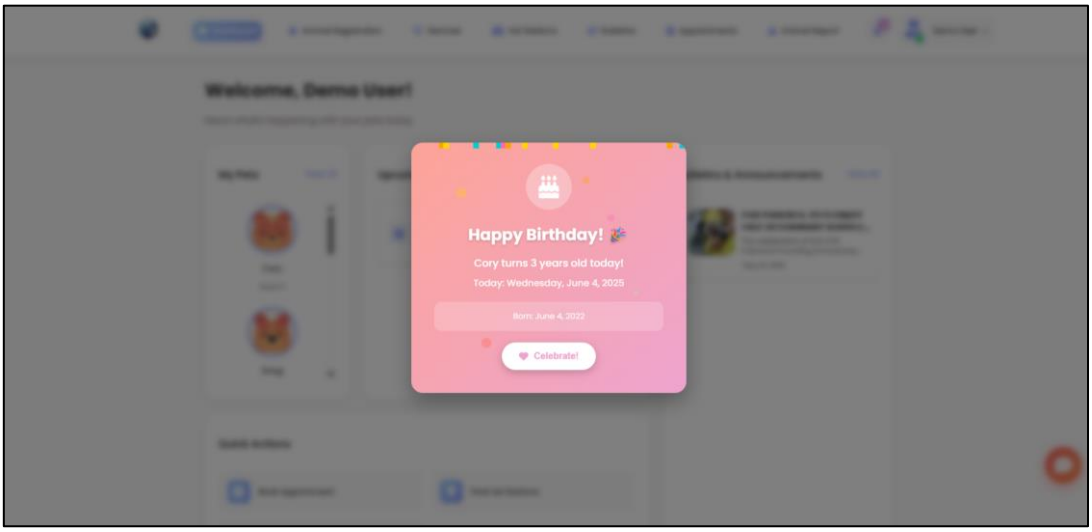

Plate 3. Selected Enhancement Integration (Web Version)

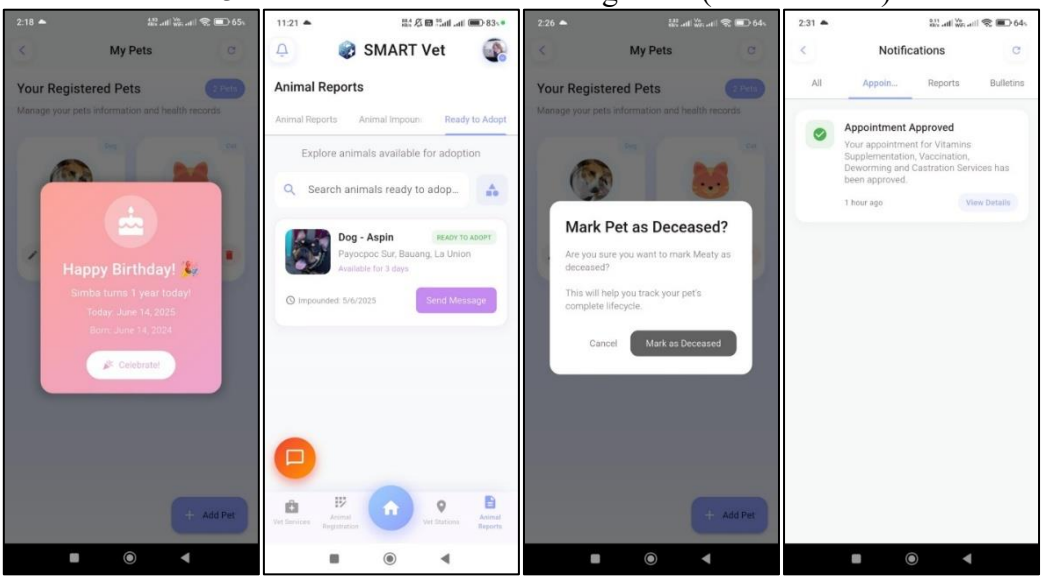

Plate 4. Selected Enhancement Integration (Mobile Version)

The acceptability analysis identified eight fundamental criteria based on focus group discussions: professional endorsement requiring multi-stakeholder approval, functional relevance evaluating platform enhancement capabilities, usability and accessibility focusing on interface simplicity, communication and collaboration capabilities, operational impact measuring service delivery improvement, implementation feasibility considering resource requirements, stakeholder satisfaction evaluating user experience quality, and continuous improvement assessing feedback integration potential.

Professional Endorsement
Definition: Acceptance from City Veterinarian officials, positive reception from private sector practitioners, and cross-sector consensus on platform value.
Theoretical Basis: This theme reflects the "social influence" component of the Unified Theory of Acceptance and Use of Technology (UTAUT) by Venkatesh et al. (2003), which emphasizes how peer opinions and professional endorsements influence technology adoption decisions.
Supporting Evidence from Focus Group Discussion Excerpts:

1. *"Maganda yung app, good luck sa study mo at sana ma-adopt ito" (The app is good, good luck with your study and I hope this gets adopted)*
2. *"As vets, we are pleased to hear na may ganitong klase ng development na nangyayari sa animal sector. Sana magtuloy ito" (As vets, we are pleased to hear that this kind of development is happening in the animal sector. I hope this continues)*
3. *"Me, si Doc Wils, Gemma, Sean, Doc Alex, and Doc Etheline kami dito sa clinic, support namin yang study mo" (Me, Doc Wils, Gemma, Sean, Doc Alex, and Doc Etheline here at the clinic, we support your study)*

Analysis: The platform demonstrates strong professional endorsement across multiple veterinary practitioners, with explicit support statements from both individual doctors and clinic teams, indicating widespread acceptance within the veterinary community. This aligns with Rogers' (2003) concept of "opinion leadership" in innovation diffusion, where endorsement from respected professionals accelerates adoption.

Functional Relevance
Definition: Platform's ability to enhance existing veterinary services, integrate with current workflows, and provide comprehensive features that address professional needs.
Theoretical Basis: This theme is grounded in Davis's (1989) "perceived usefulness" construct from TAM, which predicts that users will adopt technology if they believe it will enhance their job performance. It also reflects Sekhon et al.'s (2017) "perceived effectiveness" component of acceptability.

Supporting Evidence from Focus Group Discussion Excerpts:

1. *"Ang daming nagagawa nung application, ang ganda, sana magamit namin yan dito sa vet office namin" (The application can do so many things, it's beautiful, I hope we can use it here in our vet office)*
2. *"Mas dadali yung trabaho kung magagamit to para hindi rin magsabay sabay yung client namin, lalo magkakaron ng pagbabago sa payment para sa vaccine" (Work would be easier if we could use this so our clients wouldn't come all at once, especially there would be changes in payment for vaccines)*
3. *"Kailangan na talaga para ma-improve ang services sa City Veterinary Office namin" (This is really needed to improve the services in our City Veterinary Office)*

Analysis: The platform shows high functional relevance by addressing multiple veterinary service needs, from client management to payment processing, demonstrating comprehensive

feature coverage that enhances existing workflows. This evidence supports the task-technology fit theory by Goodhue and Thompson (1995), which suggests that technology adoption depends on how well the technology fits the tasks users need to perform.

Usability and Accessibility

Definition: Interface simplicity, multi-platform effectiveness, and technical support availability.

Theoretical Basis: This theme corresponds to Davis's (1989) "perceived ease of use" from TAM and Sekhon et al.'s (2017) "burden" component, which examines the perceived amount of effort required to participate in the intervention.

Supporting Evidence from Focus Group Discussion Excerpts:

1. "*Available ba ito sa playstore or app store?" (Is this available on the playstore or app store?)*
2. *"Nagtry po ako ngupload sa Playstore and ongoing po ang review ni Google pero available po yung app sa website just visit smartvetph.com po andun yung installer for android version" (I tried to upload it to the Playstore and Google's review is ongoing, but the app is available on the website, just visit smartvetph.com where you'll find the installer for the android version)*
3. *"Ni IT kuma ti aghandle daytoy nga application ta suda met ti agmamanage ti systems gamin awan met ti IT mi, marigatan kami agusar" (An IT person should handle this application because they are the ones who manage systems, we don't have IT staff, we'll have difficulty using it)*

Analysis: While the platform demonstrates multi-platform accessibility through web and mobile versions, concerns about technical support requirements indicate potential usability challenges for offices without IT staff. This finding aligns with digital divide theory (Norris, 2001), which highlights how technical barriers can create disparities in technology access and usage, suggesting need for enhanced user-friendly design or support systems.

Communication and Collaboration Capabilities

Definition: Information sharing effectiveness, knowledge exchange facilitation, and network building potential among veterinary professionals.

Theoretical Basis: This theme is informed by Social Network Theory (Granovetter, 1973) and Communities of Practice Theory (Wenger, 1998), which emphasize how technology can facilitate professional networks and knowledge sharing communities.

Supporting Evidence from Focus Group Discussion Excerpts:

1. *"May student din sa DMMMSU na nag-develop ng system related sa animals, pwede siguro natin i-collaborate in the future para ma-merge ang ideas" (There's also a student at DMMMSU who developed a system related to animals, maybe we can collaborate in the future to merge ideas)*
2. *"Very open ako sa collaboration, pwede naman kayo mag-approach sa akin regarding sa study niyo at willing ako mag-work together" (I'm very open to collaboration, you can approach me regarding your study and I'm willing to work together)*
3. *"Penge ng additional details para ma-feature ko sa vlog ko for promotion kapag naformal launch na" (Give me additional details so I can feature it in my vlog for promotion when it's formally launched)*

Analysis: The platform demonstrates strong potential for fostering collaboration and knowledge sharing among veterinary professionals, with evidence of networking opportunities and willingness to integrate with other systems. This supports Actor-Network Theory (Latour, 2005), which views technology adoption as a process of building networks of human and non-human actors, indicating effective communication capabilities.

Operational Impact
Definition: Service delivery improvement, administrative efficiency, and data management capabilities.
Theoretical Basis: This theme aligns with Organizational Theory perspectives on technology adoption, particularly Resource-Based View (Barney, 1991), which examines how technology resources can provide competitive advantages and improve organizational performance.

Supporting Evidence from Focus Group Discussion Excerpts:

1. *"Naglaing ka met, nagmayat diyay system, mayat kuma nu mausar mi para adda pangmanage mi ti records ditoy ayan mi" (You did well, the system is good, it would be good if we could use it so we have something to manage our records here with us)*
2. *"Maganda sana kung magamit ng city government ng San Fernando ang application na ito" (It would be good if the city government of San Fernando could use this application)*
3. *"Tamang panahon na talaga para ma-improve ang services dito sa City Vet" (It's really the right time to improve the services here at the City Vet)*

Analysis: The platform shows significant operational impact potential through improved record management, service delivery enhancement, and administrative efficiency, with stakeholders recognizing its value for government-level implementation. This evidence supports Institutional Theory (DiMaggio & Powell, 1983), which suggests that organizations adopt innovations to gain legitimacy and improve operational effectiveness.

Implementation Feasibility
Definition: Resource requirements, training needs, and maintenance sustainability to ensure long-term viability.
Theoretical Basis: This theme is grounded in Implementation Science frameworks, particularly the Consolidated Framework for Implementation Research (CFIR) by Damschroder et al. (2009), which emphasizes the importance of implementation readiness and organizational capacity.

Supporting Evidence from Focus Group Discussion Excerpts:

1. *"If gagamitin po ng city ito, need po ng Memorandum of Agreement para maindicate po dun kung ano yung extent ng help and other related matters" (If the city will use this, a Memorandum of Agreement is needed to indicate the extent of help and other related matters)*
2. *"Pwede naman po magconduct ng training para dun sa gagamit or magaadmin sa system or maghire po ng IT staff para sa office niyo po para ma-ensure ang proper handling and support" (We can conduct training for those who will use or admin the system, or hire IT staff for your office to ensure proper handling and support)*

Analysis: Implementation feasibility is addressed through structured agreements and comprehensive training programs, though concerns about IT staff requirements suggest need for careful resource planning and support infrastructure development. This aligns with Normalization Process Theory (May et al., 2009), which emphasizes the importance of organizational work required to embed new practices into routine healthcare delivery.

Stakeholder Satisfaction
Definition: User experience quality, problem-solving effectiveness, and recommendation likelihood.
Theoretical Basis: This theme reflects Sekhon et al.'s (2017) "affective attitude" component of acceptability, which captures how individuals feel about the intervention, and connects to Customer Satisfaction Theory in service management literature.

Supporting Evidence from Focus Group Discussion Excerpts:

1. *"Masaya talaga kami sa progress na nangyayari sa animal sector since wala pang ganyan na system dito sa LU, okay tong app, magamit sana" (We're really happy with the progress happening in the animal sector since there's no system like this yet here at LU, this app is okay, hope it gets used)*
2. "*Maganda to i-advertise since wala pa atang nakakagawa neto sa province" (This would be good to advertise since no one has probably made this yet in the province)*

Analysis: High stakeholder satisfaction is evident through positive feedback, enthusiasm for the platform's uniqueness in the region, and willingness to promote the system, indicating strong user experience quality and problem-solving effectiveness. This supports Expectation Confirmation Theory (Oliver, 1980), which suggests that satisfaction occurs when performance meets or exceeds expectations.

Continuous Improvement

Definition: Feedback integration, adaptability, and innovation potential, ensuring the platform can evolve with changing veterinary service needs.

Theoretical Basis: This theme is informed by Adaptive Capacity Theory (Walker et al., 2004) and Innovation Management Theory (Tidd & Bessant, 2020), which emphasize the importance of continuous learning and adaptation in technology development and implementation.

Supporting Evidence from Focus Group Discussion Excerpts:

1. Collaborative opportunities for system enhancement and idea merging were mentioned.
2. Researcher's openness to collaboration and willingness to work together on improvements.
3. Discussion of future collaboration with other development projects in the field.

Analysis: The platform demonstrates strong continuous improvement potential through developer openness to feedback, collaborative development approaches, and integration possibilities with other veterinary systems, ensuring long-term adaptability and innovation. This aligns with Dynamic Capabilities Theory (Teece et al., 1997), which emphasizes organizations' ability to integrate, build, and reconfigure internal and external competences to address rapidly changing environments.

The platform received universal endorsement from veterinarians across both public and private sectors, with participants expressing enthusiasm for adoption and willingness to promote the system, while addressing multiple veterinary service needs including client management, payment processing, record keeping, and service delivery optimization. However, technical infrastructure gaps present barriers as many veterinary offices lack dedicated IT support and staff require additional training to effectively utilize the system, though the SWOT analysis reveals significant strengths including universal stakeholder approval, comprehensive feature set, strong institutional backing, and clear market readiness, positioning SMART Vet with strong market potential and first-mover advantage in an underserved regional market while success depends on addressing technical support requirements and infrastructure limitations for sustainable long-term implementation.

SMART Vet Platform SWOT Analysis

STRENGTHS	WEAKNESSES
• Zero negative feedback from all participants • High approval from both LGU and private sector veterinarians • Effectively addresses current veterinary service needs • Strong support from LGU leadership • Private sector willingness to actively promote platform • Veterinary community shows genuine eagerness for modernization • Strong cooperation among stakeholders • All user groups demonstrate understanding of platform benefits	• Veterinary office lacks dedicated IT support • Staff possess insufficient technical skills • Potential reliance on external technical support • Staff need additional training • Current technical capabilities may not match platform requirements • Comprehensive training required before full deployment • •
• **OPPORTUNITIES**	• **THREATS**
• Collaborative development with academic institutions • Potential MOA with city government • Opportunities to expand networks • Integration with related animal health systems • Catalyst for improved veterinary services • Broader regional implementation possibilities • Foundation for citywide animal healthcare modernization • Potential to become model for other municipalities • Potential to attract additional funding and support	• Success dependent on securing formal MOA • Potential delays if agreements not finalized • Ongoing challenges in maintaining support • Risk of underutilization due to inadequate training • Potential issues without dedicated IT support • Reliance on sustained institutional support • Financial limitations affecting implementation • Threats from competitive solutions • Risks from technology-related dependencies

Conclusion

This research validates the transformative potential of the SMART Vet platform as a cornerstone of veterinary digital transformation and public health advancement. The study's key findings demonstrate that locally developed government applications can achieve exceptional international quality standards, with SMART Vet attaining an ISO/IEC 25010 rating of 4.49 ("Excellent and Very Highly Usable"). The platform's successful integration of four core features through user-centered development and unanimous professional endorsement confirms its readiness for widespread implementation, representing a significant contribution to digitizing veterinary services and enhancing public health surveillance capabilities. The SMART Vet knowledge management platform application addresses critical

gaps in veterinary service delivery by providing comprehensive knowledge management tools that can revolutionize how Local Government Units approach animal health monitoring, disease prevention, and public health protection. This digital transformation extends beyond operational efficiency to create robust data ecosystems that support evidence-based decision-making and facilitate rapid response to veterinary emergencies and zoonotic disease threats.

Moving forward, sustained expansion requires strategic pilot implementation across diverse LGU contexts to validate scalability and adaptability. Longitudinal studies spanning 3-5 years are essential to measure long-term impact on veterinary service outcomes, user adoption patterns, and public health indicators. The research framework established through this study provides a replicable model for nationwide veterinary digitization, positioning SMART Vet as the foundation for systematic transformation of government veterinary services.

However, realizing these potential demands sustained institutional support and robust technical infrastructure development. Success hinges on establishing regional technical support centers, implementing mandatory IT capacity building programs, and securing dedicated budget allocations for continuous platform enhancement. Without this comprehensive support ecosystem, even the most innovative digital solutions risk implementation failure and limited sustainability.

The study recommends immediate institutionalization of the SMART Vet development framework as the standard for LGU digital governance initiatives, with ISO/IEC 25010 compliance as the mandatory quality benchmark. This research contributes significantly to the broader veterinary digital transformation agenda while demonstrating that public sector innovation can achieve world-class standards when supported by rigorous development methodologies and sustained institutional commitment. The future of veterinary public health depends on embracing such comprehensive digital solutions with the infrastructure and support necessary to ensure their long-term success and scalability.

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Statement of Conflict of Interest

The authors declare no commercial or financial conflicts of interest in this research. The lead researcher developed the SMART Vet platform as part of an academic study and maintained independence throughout the evaluation process. While having a vested interest in demonstrating the platform's effectiveness, the evaluation employed standardized ISO/IEC 25010 quality characteristics and involved independent stakeholders including LGU veterinarians and private practitioners in focus group discussions to ensure objective assessment. The research was conducted for educational purposes without commercial intent or external funding. All enhancements were based on user recommendations rather than commercial considerations. The authors acknowledge their dual role as developers and

evaluators but emphasize that the assessment utilized established international standards and multi-stakeholder validation to maintain research integrity and minimize evaluation bias.

References

Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, *50*(2), 179-211.

Anti-Red Tape Authority. (2022). *The Ease of Doing Business Law*. https://arta.gov.ph/about/the-eodb-law/

Babbie, E. R. (2020). *The practice of social research* (15th ed.). Cengage Learning.

Barbour, R. S. (2014). *Introducing qualitative research: A student's guide* (3rd ed.). SAGE Publications.

Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, *17*(1), 99-120.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77-101.

Buchanan, J. M., & Tullock, G. (1962). *The calculus of consent: Logical foundations of constitutional democracy*. University of Michigan Press.

Bunge, M. (1997). Mechanism and explanation. *Philosophy of the Social Sciences*, *27*(4), 410-465.

Centre for Public Impact. (2024). *The Anti-Red Tape Act in the Philippines*. https://centreforpublicimpact.org/public-impact-fundamentals/the-anti-red-tape-act-in-the-philippines/

City Government of San Fernando. (2002). *City Ordinance No. 2002-009: An Ordinance regulating the maintenance and safekeeping of animals for the welfare and protection of residents in the 59 Barangays of the City of San Fernando*. San Fernando, Philippines.

City Government of San Fernando. (2007). *City Ordinance No. 2007-005: The Animal Welfare Code of the City of San Fernando*. San Fernando, Philippines.

City Government of San Fernando. (2008). *City Ordinance No. 2008-006: Anti-Stray Animal Ordinance*. San Fernando, La Union, Philippines.

City of San Fernando, La Union. (2025). *State of the City Address Technical Report 2025*. City Government of San Fernando, La Union.

City of San Fernando, La Union City Veterinary Office. (2025). *2025 office latest reports*. City Government of San Fernando, La Union.

Creswell, J. W., Plano Clark, V. L., Gutmann, M. L., & Hanson, W. E. (2018). *Research design: Qualitative, quantitative, and mixed methods approaches* (5th ed.). SAGE Publications.

Damschroder, L. J., Aron, D. C., Keith, R. E., Kirsh, S. R., Alexander, J. A., & Lowery, J. C. (2009). Fostering implementation of health services research findings into practice: A consolidated framework for advancing implementation science. *Implementation Science*, *4*(1), 50.

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, *13*(3), 319-340.

Department of Agriculture. (2024). *Philippines' farm sector rose at a faster clip in 2023 on poultry, livestock gains, record rice output*. https://www.da.gov.ph/philippines-farm-sector-rose-at-faster-clip-in-2023-on-poultry-livestock-gains-record-rice-output

Department of Information and Communications Technology. (2020). *E-Government harmonization program*. https://dict.gov.ph

Department of Information and Communications Technology. (2022). *E-Government Masterplan 2022*. https://dict.gov.ph/ictstatistics/wp-content/uploads/2019/07/EGMP_Book_Abridged.pdf

Department of the Interior and Local Government. (2021). *E-Government systems in local government units*.

Department of Trade and Industry Philippines. (2025). *Data Privacy in DTI*. https://www.dti.gov.ph/good-governance-program/data-privacy-dti/

Digital Watch Observatory. (2024). *Philippine Digital Transformation Strategy 2022*. https://dig.watch/resource/philippine-digital-transformation-strategy-2022

DiMaggio, P. J., & Powell, W. W. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, *48*(2), 147-160.

Dyson, R. G. (2004). Strategic development and SWOT analysis at the University of Warwick. *European Journal of Operational Research*, *152*(3), 631-640.

FAO. (2020). *EMPRES-i: Enhancing disease surveillance*. Food and Agriculture Organization of the United Nations.

FAO. (2023). *EMPRES-i: Global Animal Disease Information System*. https://empres-i.fao.org

Faulkner, L. (2003). Beyond the five-user assumption: Benefits of increased sample sizes in usability testing. *Behavior & Information Technology*, *22*(5), 355-364. https://doi.org/10.1080/01449290310001649359

Fernando, J. (2023, February 3). Las Piñas LGU launches Veterinary Office IT Management System. *Manila Bulletin*. https://mb.com.ph/2023/2/3/las-pinas-lgu-launches-veterinary-office-it-management-system#google_vignette

Gandhi, M. (n.d.). *The greatness of a nation and its moral progress can be judged by the way its animals are treated*. https://www.goodreads.com

Garcia, L., Perez, M., & Torres, D. (2021). Continuous quality improvement in public health services: Integrating feedback mechanisms for better outcomes. *Public Health Management Review*, *29*(4), 52-67.

Goodhue, D. L., & Thompson, R. L. (1995). Task-technology fit and individual performance. *MIS Quarterly*, *19*(2), 213-236.

Granovetter, M. S. (1973). The strength of weak ties. *American Journal of Sociology*, *78*(6), 1360-1380.

Heeks, R. (2003). Most eGovernment-for-development projects fail: How can risks be reduced? *Government Information Quarterly*, *20*(4), 321-334.

Hood, C. (1991). A public management for all seasons? *Public Administration*, *69*(1), 3-19. https://doi.org/10.1111/j.1467-9299.1991.tb00779.x

International Organization for Standardization. (2011). *ISO/IEC 25010:2011 - Systems and software engineering — Systems and software Quality Requirements and Evaluation*

(SQuaRE) — System and software quality models. ISO. https://www.iso.org/standard/35733.html

Journal of Animal Health and Production. (2020). The impact of digital technologies on veterinary services and disease surveillance. *Journal of Animal Health and Production*, *38*(2), 123-130.

Journal of Global Health. (2021). Digital surveillance in veterinary services: Impacts on zoonotic disease management.

Latour, B. (2005). *Reassembling the social: An introduction to actor-network-theory*. Oxford University Press.

LawPhil. (1998). *Republic Act No. 8485: The Animal Welfare Act of 1998*. https://lawphil.net/statutes/repacts/ra1998/ra_8485_1998.html

Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology*, *140*, 1-55.

Marakas, G. M. (2003). *Decision support systems in the 21st century* (2nd ed.). Pearson Prentice Hall.

May, C., Mair, F., Finch, T., MacFarlane, A., Dowrick, C., Treweek, S., Rapley, T., Ballini, L., Ong, B. N., Rogers, A., Murray, E., Elwyn, G., Légaré, F., Gunn, J., & Montori, V. M. (2009). Development of a theory of implementation and integration: Normalization process theory. *Implementation Science*, *4*(1), 29.

Miles, M. B., & Huberman, A. M. (1994). *Qualitative data analysis: An expanded sourcebook* (2nd ed.). Sage Publications.

Morgan, D. L. (1998). The focus group interview. In D. L. Morgan (Ed.), *Planning focus groups* (pp. 3-15). Sage Publications.

National Privacy Commission. (2024). *Republic Act 10173 - Data Privacy Act of 2012*. https://privacy.gov.ph/data-privacy-act/

Neuman, W. L. (2014). *Social research methods: Qualitative and quantitative approaches* (7th ed.). Pearson.

Nielsen, J. (2000). *Why you only need to test with 5 users*. Jakob Nielsen's Alertbox. https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/

Nonaka, I., & Takeuchi, H. (1995). *The knowledge-creating company: How Japanese companies create the dynamics of innovation*. Oxford University Press.

Norris, P. (2001). *Digital divide: Civic engagement, information poverty, and the Internet worldwide*. Cambridge University Press.

Official Gazette of the Republic of the Philippines. (2016). *Republic Act No. 10844 - An Act Creating the Department of Information and Communications Technology*. https://www.officialgazette.gov.ph/2016/05/23/republic-act-no-10844/

OIE. (2021). *WAHIS: A tool for animal disease reporting*. World Organisation for Animal Health.

Oliver, R. L. (1980). A cognitive model of the antecedents and consequences of satisfaction decisions. *Journal of Marketing Research*, *17*(4), 460-469.

OpenGov Asia. (2019). *The Philippines' DICT launches E-Government Masterplan 2022*. https://opengovasia.com/2019/06/24/the-philippines-dict-launches-e-government-masterplan-2022/

OpenGov Asia. (2024). *DICT empowering digital excellence in the Philippines*. https://opengovasia.com

Palinkas, L. A., Horwitz, S. M., Green, C. A., Wisdom, J. P., Duan, N., & Hoagwood, K. (2015). Purposeful sampling for qualitative data collection and analysis in mixed method implementation research. *Administration and Policy in Mental Health and Mental Health Services Research*, *42*(5), 533-544. https://doi.org/10.1007/s10488-013-0528-y

Philippine Statistics Authority. (2020). *Projected population for the City of San Fernando, La Union*. https://openstat.psa.gov.ph

Philippine Statistics Authority. (2024). *Livestock and poultry statistics in the Philippines*. https://rssocar.psa.gov.ph/livestock-poultry

Pickton, D. W., & Wright, S. (1998). What's SWOT in strategic analysis? *Strategic Change*, *7*(2), 101-109.

Pressman, R. S. (2014). *Software engineering: A practitioner's approach* (8th ed.). McGraw-Hill Education.

Republic of the Philippines. (1991). *Republic Act No. 7160: An Act providing for a Local Government Code of 1991*. Official Gazette. https://www.officialgazette.gov.ph/1991/10/10/republic-act-no-7160/

Republic Act No. 10844, Department of Information and Communications Technology Act of 2015. (2016). https://www.officialgazette.gov.ph/2016/05/23/republic-act-no-10844/

Robbins, S. P., & Coulter, M. (2018). *Management* (14th ed.). Pearson.

Robson, C. (2011). *Real World Research: A Resource for Social Scientists and Practitioner-Researchers*. Wiley.

Rogers, E. M. (2003). *Diffusion of innovations* (5th ed.). Free Press.

Securiti. (2025). *Philippines Data Privacy Act's Application to AI Systems Processing Personal Data*. https://securiti.ai/philippines-data-privacy-act-application-to-ai-systems-processing-personal-data/

SecurePrivacy. (2024). *Data Privacy Act 2012 | Rights and Compliance in the Philippines*. https://secureprivacy.ai/blog/data-privacy-act-2012-philippines-guide

Sekhon, M., Cartwright, M., & Francis, J. J. (2017). Acceptability of healthcare interventions: An overview of reviews and development of a theoretical framework. *BMC Health Services Research*, *17*(1), 88.

Sommerville, I. (2016). *Software engineering* (10th ed.). Pearson.

Studocu. (2024). *E-Government Master Plan 2022, A Digitized Government for the Nation*. https://www.studocu.com/ph/document/lyceum-of-the-philippines-university/ab-foreign-service/e-government-master-plan-2022-a-digitized-government-for-the-nation/19394460

Taherdoost, H. (2022). Determination of Business Strategies Using SWOT Analysis; Planning and Managing the Organizational Resources to Enhance Growth and Profitability. *Available at SSRN*.

Teddlie, C., & Tashakkori, A. (2009). *Foundations of Mixed Methods Research: Integrating Quantitative and Qualitative Approaches in the Social and Behavioral Sciences*. SAGE Publications.

Teece, D. J., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, *18*(7), 509-533.

Tidd, J., & Bessant, J. (2020). *Managing innovation: Integrating technological, market and organizational change* (7th ed.). John Wiley & Sons.

U.S. Food and Drug Administration. (2024). *Animal and Veterinary Innovation Agenda*. https://www.fda.gov/animal-veterinary/products/animal-and-veterinary-innovation-agenda

Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. *Management Science*, *46*(2), 186-204.

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, *27*(3), 425-478.

Wagner, S. (2013). Assessing functional appropriateness in software systems. *Proceedings of the International Conference on Software Quality*, 45-54. https://doi.org/10.1109/ICSQ.2013.6543210

Walker, B., Holling, C. S., Carpenter, S. R., & Kinzig, A. (2004). Resilience, adaptability and transformability in social–ecological systems. *Ecology and Society*, *9*(2), 5.

Wang, L., & Zhang, X. (2020). Designing and evaluating educational programs: A developmental research approach. *Educational Research Quarterly*, *43*(2), 123-138.

Wenger, E. (1998). *Communities of practice: Learning, meaning, and identity*. Cambridge University Press.

Wiig, K. M. (1993). *Knowledge management foundations: Thinking about thinking – How people and organizations create, represent, and use knowledge*. Schema Press.

WOAH. (2024). *World Animal Health Information System (WAHIS)*. https://www.woah.org/en/what-we-do/animal-health-and-welfare/disease-data-collection/world-animal-health-information-system/

World Organisation for Animal Health (WOAH). (2021). *Veterinary paraprofessional workshop – Philippines*. rr-asia.woah.org